

Clinical Trial Ops & TMF

A Governed Agentic AI Reference Architecture for Life Sciences

Agent 03 of 8 · surface TMF gaps and EDC query backlog continuously — a CRA approves every disposition

HCLS AI Agent Suite

HCLS AI Agent Suite · Built on AWS · June 2026

FROM LATE SURPRISES TO CONTINUOUS READINESS

Clinical Trial Ops & TMF

A governed trial-ops copilot that monitors TMF completeness and EDC query backlog continuously and drafts the work — while a CRA or TMF owner approves every filing, query, and closure, keeping the trial inspection-ready by design.

~\$800K / day

lost/delayed sales per day of trial delay (+
~\$40K/day direct cost)

[gov/peer-reviewed — Tufts 2024]

57%

of TMF owners still rely on paper / simple e-
file systems

[industry-research — Veeva]

\$19.0M

median pivotal-trial cost (IQR \$12.2M–
\$33.1M)

[gov/peer-reviewed — JAMA IM]

The Issue & The Cost of Doing Nothing

THE ISSUE

- TMF gaps surface late — often after database lock, when they are most expensive to fix.
- EDC query backlogs slow enrollment and data cleaning; completeness is continuous, not periodic.
- “Failure to maintain adequate/accurate case histories” is among the most common FDA BIMO 483 findings.
- Advanced-eTMF users report inspection readiness of 56% vs. 25% — tooling gap is measurable.

THE COST OF DOING NOTHING

~\$25.7M / slip

Modeled: a 30-day database-lock / TMF slip on one Phase III $\approx 30 \times (\$800K \text{ lost sales} + \$55,716 \text{ direct/day}) \approx \sim \$25.7M$. Tufts 2024 supersedes the old “\$4-8M/day” figure.

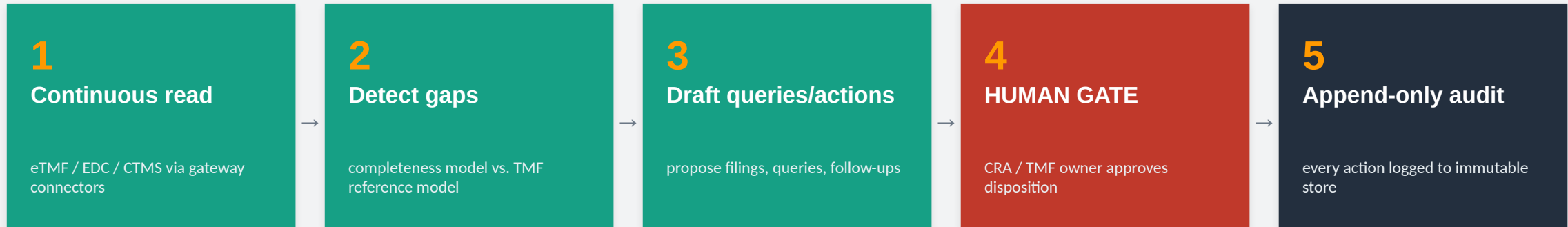
- Inspection findings and refuse-to-file risk from an incomplete TMF.
- Enrollment and lock delays compounding the per-day cost of delay.

[modeled — Tufts 2024 day-of-delay $\times$ Phase III direct cost]

The design line: the agent detects gaps and drafts queries; a CRA or TMF owner approves every filing, query, and closure — the disposition is human-owned.

How We Solve It — A Governed Pipeline

Continuously read eTMF/EDC/CTMS through a scoped gateway, detect gaps and draft queries — then route every disposition to a human gate.



Every action audited

node, timestamp, actor, data sources, model + prompt version — append-only and inspection-ready (21 CFR Part 11).

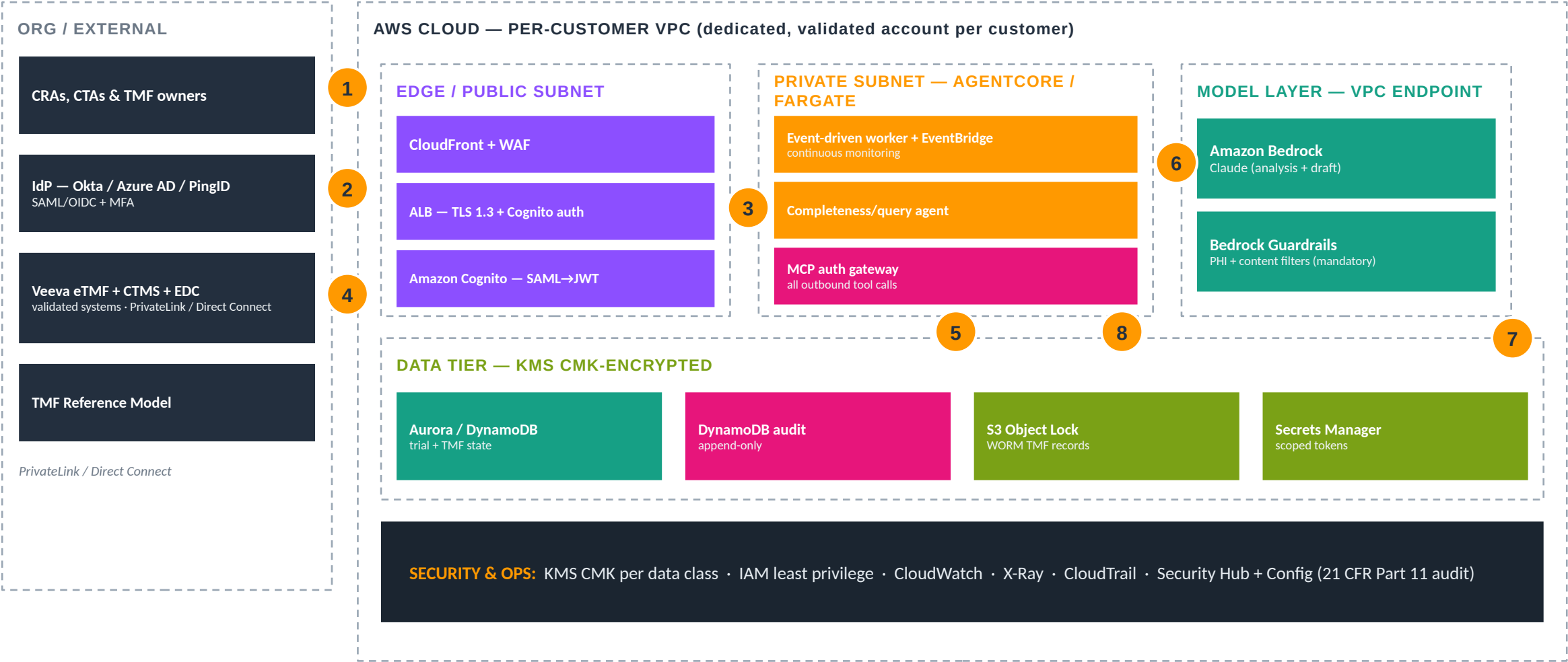
Grounded & explainable

every regulated claim traces to the source corpus; Bedrock Guardrails mask PHI before any model call and bound the output.

AI never decides the TMF/query disposition

the consequential action is a human-only step, enforced in the orchestration framework — not a policy SOP.

AWS Architecture & Traffic Flow



1 user sign-in 2 SAML federation (MFA at IdP; Cognito issues JWT) 3 authenticated session to trial-ops console 4 eTMF/EDC/CTMS reads over private connectivity 5 EventBridge schedules continuous completeness checks 6 model calls stay in-VPC via Bedrock + Guardrails 7 every gap, query, and disposition persisted to append-only audit 8 connectors reachable only through the governed MCP gateway

Proof, Payback & How to Deploy

MEASURED OUTCOMES

~\$800K/day

delay cost the agent works to compress

[gov/peer-reviewed — Tufts]

1M+ docs

auto-classified by a TMF bot, “tens of thousands of hours”

[vendor — Veeva]

56% vs 25%

inspection readiness, advanced-eTMF vs. manual

[industry-research — Veeva]

continuous

completeness vs. periodic spot-checks today

[design property]

WHAT IT TAKES TO DEPLOY

1. Provision KMS CMK + per-customer validated VPC / network.
1. Stand up Cognito SAML→JWT federation (CRA / CTA / TMF-owner roles).
1. Front with CloudFront + WAF, ALB TLS 1.3.
1. Deploy the agent + grant tools via the MCP gateway (deny-by-default).
1. Connect Veeva eTMF + CTMS + EDC; load the TMF Reference Model.
1. Wire EventBridge for continuous checks; enable audit; run smoke + HITL test.

Deploy: CloudFormation quick-deploy provisions the isolated environment + dual MCP gateway + event-driven worker; point connectors at eTMF/EDC/CTMS.

aws-native-reference/03-clinical-trial-ops/DEPLOY.md · docs/DEPLOY-QUICKSTART.md

The takeaway: the agent isn't the product — the governed platform that makes it 21 CFR Part 11 / GxP-defensible and deployable on AWS is.